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U–Pb baddeleyite ages and Hf, Nd isotope chemistry constraining repeated mafic magmatism in the Fennoscandian Shield from 1.6 to 0.9 Ga

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Abstract The Palaeoproterozoic (1.90–1.60 Ga) crust of central Fennoscandia was intruded repeatedly by dolerite dikes and sills during the Neo- and Mesoproterozoic eons. We report 17 new baddeleyite U–Pb dates comprising six generations of dolerites (in Ma):

Blekinge-Dalarna dolerites	946–978
Protogine Zone dolerites	1,215–1,221
Central Scandinavian Dolerite Group	1,264–1,271
Tuna dikes and age equivalents in Dalarna	1,461–1,462
Värmland dolerites	~1,568
Breven-Hällefors dolerites	~1,595

The favoured tectonic model implies that the majority of these suites were related to active margin processes somewhere west (and possibly south) of the Fennoscandian Shield. Dolerite intrusions are interpreted to reflect discrete events of back-arc extension as the arc retreated oceanward. Initial Hf and Nd isotope compositions of the dolerite swarms fall between CHUR and normal-depleted mantle, and suggest a variably depleted and re-enriched mantle as the source for the here investigated 1.6 to 0.95 Ga old mafic rocks. Repeated recycling of older crustal components, mainly sediments (dominated by material with short residence ages) in earlier subduction systems may have been very efficient at producing geochemically and isotopically variably enriched lithospheric mantle sections beneath the Fennoscandian Shield.

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Introduction

This study focuses on Meso- to Neoproterozoic dolerite dikes and sills in Sweden and Finland, east of the SFDZ (Sveconorwegian Frontal Deformation Zone) and the PZ (Protogine Zone) (Fig. 1). Accurate and precise ages of mafic rocks are important for many reasons. First, dolerites and large igneous province (LIP) basalts represent ancient events of extension that in some cases lead to break-up of continents (e.g. Fahrig 1987; Ernst et al. 2005). Second, mafic rocks are efficient in recording and preserving the declination and inclination of the Earth's magnetic field from the time the rock was formed (e.g. Bylund and Pesonen 1987). These qualities make mafic rocks key geological markers for correlating and reconstructing the ancient positions of lithospheric plates. Third, the timing and the regional occurrence of most dolerite suites in Scandinavia are known only to a first order based on Rb–Sr (Klingspor 1976; Patchett 1978; Johansson and Johansson 1990), Sm–Nd (Johansson and Johansson 1990) and the K–Ar (Klingspor 1976) isotope methods. Although these methods, particularly if applied with a single-sample approach, can provide accurate emplacement ages, there are situations where not all isochron criteria (i.e. closed system

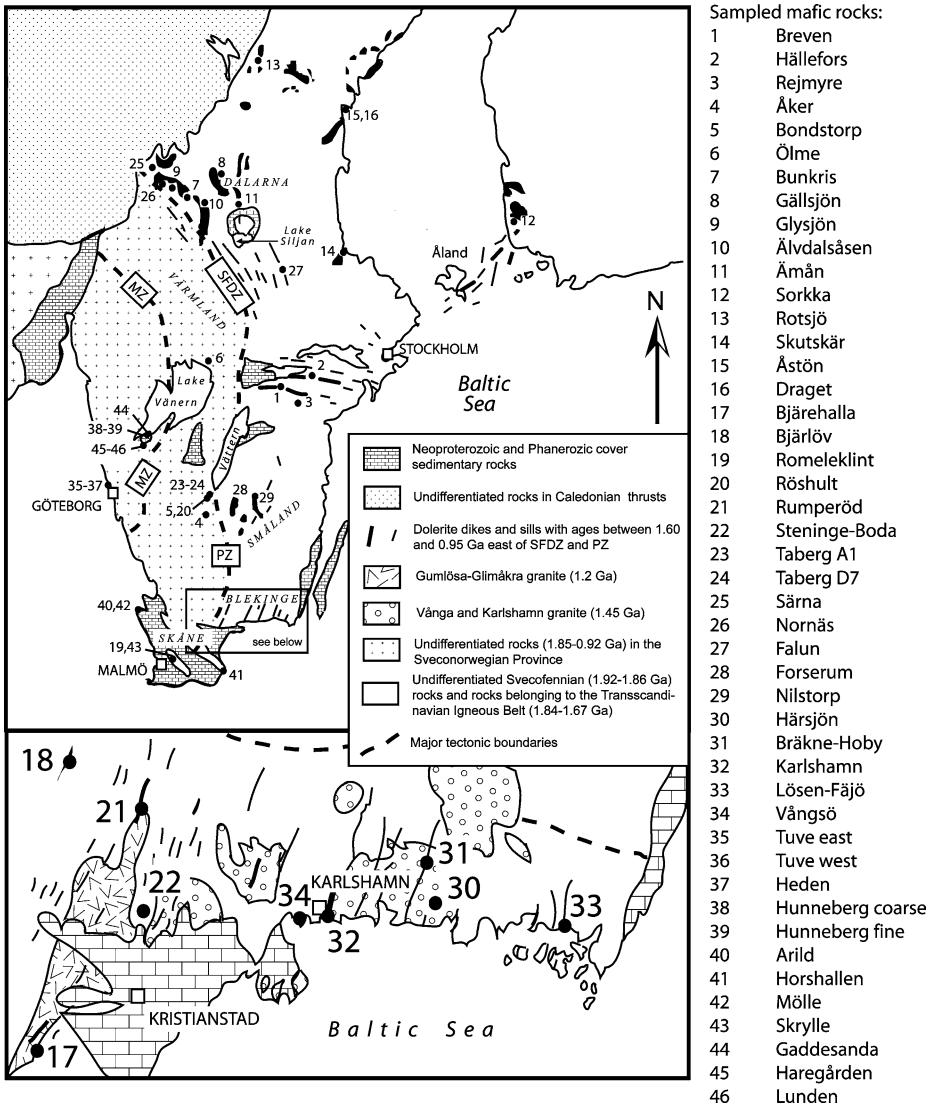


Fig. 1 Simplified geological maps of southern Sweden. *Numbers* refer to sample locations listed to the *right*. The *lower* figure magnifies framed area in *upper* figure. The area covers most of Blekinge and the northeastern corner of Scania. *MZ* Mylonite Zone; *PZ* Protogine Zone; *SFDZ* Sveconorwegian Frontal Deformation Zone (modified after Johansson and Johansson 1990 and Patchett 1978)

behaviour, uniform initial daughter element ratios, etc.) are fulfilled and the results are geologically meaningless (cf. Romer and Yilin 2005). Many of these limitations are largely irrelevant in U–Pb chronology because the minerals usually employed for dating of mafic rocks (e.g. zircon and baddeleyite) often have high closure temperatures (>900°C), because closed-system behaviour is easily evaluated (i.e. by comparing $^{207}Pb/^{235}U$ with $^{206}Pb/^{238}U$ dates), and because it can be applied to single minerals (i.e. independence from the issue of isotopic equilibrium between minerals in a sample).

Baddeleyite (ZrO_2) crystallizes directly from a mafic magma, never occurs as xenocrysts, contains modest to high amounts of uranium (>200 ppm) with negligible common lead, and so it is the preferred mineral for U–Pb geochronology of mafic rocks (e.g. Heaman and LeCheminant 1993). The development of new efficient mineral separation techniques to extract baddeleyite grains from dolerite samples (Söderlund and Johansson 2002) now allows the possibility to date reliably large number of mafic samples. We report baddeleyite U–Pb dates for 17 dolerite dikes and sills belonging to six different generations in the range 1.6–0.95 Ga. We also present initial Hf and Nd isotope compositions for these rocks and 35 additional samples and discuss plausible mantle sources and tectonic settings for Proterozoic mafic magmatism in Fennoscandia within the ~650 m.y. time interval of dolerite intrusion.

Geological setting

The Fennoscandian Shield is divided into different lithotectonic belts with the oldest rocks (Archaean and Palaeoproterozic) in the northeast. The south-central parts of the Fennoscandian Shield are dominated by Palaeo- to early Mesoproterozoic crust originally accreted during the early Svecofennian (1.91–1.86 Ga), that later was intruded by the voluminous 1.85–1.65 Ga Transscandinavian Igneous Belt (TIB; cf. Högdahl et al.

2004) (Fig. 1). The TIB is made up of different generations of granitoid batholiths, with associated basic and volcanic rocks, trending across the Scandinavian peninsula in an approximately N-S direction. Present age data from south Norway and southwest Sweden indicate that crustal growth continued through juvenile arc magmatism along the western margin of the shield between 1.66 and 1.50 Ga (e.g. Å häll et al. 2000; Andersen et al. 2004), but from c. 1.6 Ga essentially all crust in central Fennoscandia was stable and only minor additions of crustal material took place subsequently. It is in this tectonic environment that all mafic events of this study took place. The major structural and metamorphic characteristics and the overall tectonic architecture of Proterozoic rocks in south Norway and western Sweden were acquired during the Sveconorwegian (1.1–0.95 Ga; e.g. de Haas et al. 1999; Bingen et al. 2001) and Caledonian (0.6–0.4 Ga) orogenies.

Overview of Proterozoic and Palaeozoic dolerites in central Fennoscandia

The Breven-Hällefors dolerites (c 160 Ga)

A system of E–W trending, tholeiitic to mildly alkali-enriched dolerites occurs in a 120×60 km area between Lake Vättern and Stockholm (Fig. 1). The Breven and the Hällefors dikes are by far the largest, about 1 km in width and some tens of km in length. They are composite, zoned intrusions. Thinner dikes of this suite typically carry plagioclase phenocrysts. Rb–Sr mineral isochron dates of the Breven and Hällefors dikes are 1,542 ± 23 and 1,518 ± 38 Ma, respectively (Patchett 1978). The petrography, geochemistry and emplacement history of the Breven-Hällefors dikes are reported by e.g. Gorbatschev et al. (1987).

The Värmland dolerites (c 1.57 Ga)

The so called Värmland dolerites (previously termed 'hyperites') occur as sheared, partly amphibolitized sheets in the area north of Lake Vänern (Fig. 1). The degree of metamorphism varies from perfectly preserved igneous mineralogy and texture to garnet amphibolites. Johansson and Johansson (1990) reported a Sm–Nd mineral isochron age of 1,512 ± 98 Ma for a fresh dolerite sample. The U–Pb baddeleyite date of ~1,568 Ma (Wahlgren et al. 1996) for two dolerite dikes immediately east of the SFDZ (Sveconorwegian Frontal Deformation Zone; Wahlgren et al. 1994) is probably so far the best age estimate for this generation of dolerites. Ductile deformation and amphibolite facies metamorphism occurred in this area at 976–956 Ma based on U–Pb titanite ages (Söderlund et al. 1999) and cooling below the muscovite Ar blocking temperature occurred at 930–905 Ma based on ^{40}Ar–^{39}Ar white mica ages (Page et al. 1996). The mineralogical and geochemical characteristics of the dolerites in Värmland have been described by e.g. Morthorst et al. (1983).

New radiometric data of partly metamorphosed mafic intrusions along the PZ south of Lake Vättern (Fig. 1) have been obtained recently. Zircons in the Å ker metabasite have been dated by two techniques at 1,565 ± 5 (Pb–Pb evaporation; Ask 1996) and 1,562 ± 6 Ma (U–Pb SIMS; P. Söderlund et al. 2004a). Even further to the south, a metadolerite has given a baddeleyite U–Pb age of 1,574 ± 9 Ma (C. Möller, personal communication). These results show that mafic intrusions of this generation extend far outside and south of the Värmland region.

The Tuna dolerites and age equivalents (c 147–146 Ga)

In southern Dalarna (Fig. 1), a distinct swarm of at least several hundred dikes are trending in a northeasterly direction. These are typically < 1 m wide, fine-grained and carrying abundant plagioclase megacrysts. Patchett (1978) reported a Rb–Sr whole-rock age of 1,388 ± 50 Ma for dikes of this swarm. The dolerites are associated, and sometimes mingled, with quartz-porphyries (the Gustafs porphyries) essentially of granitic compositions, dated at 1,474 ± 4 Ma (Lundström et al. 2002). The alkaline Noran pluton was dated at 1,469 ± 10 Ma (Claesson and Kresten 1997). A large N-trending dike at Bunkris in northern Dalarna yielded Rb–Sr dates as old as 1,516 ± 62 and 1,546 ± 84 Ma. Patchett (1978) discussed the likelihood that these dates might be biased towards too old ages due to the effects of crustal contamination. If this is correct, the dikes in the north may be coeval with the Tuna dikes in the south. All in all, Rb–Sr data suggest that ~1.4 Ga dolerites co-exist with at least two younger suites of mafic dolerites in Dalarna, and that ~1.4 Ga dolerites occurs both north and south of Lake Siljan (Fig. 1).

The Västergötland dolerites (c 1.3 Ga)

This dolerite suite was discovered only a few years ago during regional mapping by the Geological Survey of Sweden. Dikes have been observed in an area immediately south of Lake Vänern near the late Palaeozoic dolerite sills at Halle- and Hunneberg (Fig. 1). Due to limited exposures, the number of dikes and their regional importance are difficult to assess. The dominating dike direction is NNE. The most characteristic feature of the dikes is the imprint of granulite facies metamorphism. Baddeleyite grains in a dolerite sample with well preserved igneous texture indicate an age around 1300 Ma for this swarm, whereas U–Pb, Lu–Hf and Sm–Nd isotope chronology date the event of high-pressure granulite metamorphism at 1,045–1,025 Ma (U. Söderlund et al., in review). The dike swarm is located in the crustal segment west of the Mylonite Zone, where it co-exists with equally old granite intrusions.

The Central Scandinavian Dolerite Group (CSDG) (c 127–126 Ga)

The Central Scandinavian Dolerite Group (CSDG) is made up primarily of sill-like bodies, often hundreds of metre in thickness (e.g. Gorbatschev et al. 1979, 1987). The occurrence of CSDG stretches over wide areas from northern Dalarna to Västerbotten (Fig. 1), and covers a total area of about 100,000 km^2 in Sweden and western Finland (Satakunta). Elming and Mattsson (2001) divided CSDG into five different complexes based on their regional distribution (see Fig. 1 in their paper). CSDG are made up of mildly alkaline rocks classified as olivine diabases (e.g. Suominen 1991; Solyom et al. 1992; Patchett et al. 1994). Patchett (1978) reported Rb–Sr WR-mineral isochron dates for eight dolerite sills in the Dalarna-Jämtland region, ranging in age between 1,156 and 1,290 Ma, with Rb–Sr WR-biotite ages clustered between 1,213 and 1,247 Ma. U–Pb zircon and baddeleyite dates of four large sill complexes in Finland gave robust baddeleyite U–Pb dates between 1,258 and 1,268 Ma (Suominen 1991). The baddeleyite U–Pb age (1,258 ± 10 Ma) for one of the samples dated by Suominen (1991), the Sorkka sill, was reproduced by Söderlund et al. (2004b; 1256.2 ± 1.4 Ma).

The Protogine Zone dolerites (c 122 Ga)

Johansson and Johansson (1990) presented radiometric evidences for the presence of a second set of dolerites along the PZ. The Röshult and Rumperöd dikes were dated at 1,184 ± 98 and 1,177 ± 38 Ma by the Sm–Nd isotopic method applied to magmatic mineral fractions and whole rock powder of well-preserved dolerite samples. These ~1.2 Ga dolerites are typically represented by dikes that are thicker (> 20 m) than dikes and sills that belong to BDD (next chapter). They are remarkably dark as the result of submicroscopic Fe-bearing mineral inclusions in magmatic plagioclase (Estifanos et al. 1997). Typically, the centres of the dikes show a well-preserved magmatic mineralogy and texture that becomes progressively transformed into a ductile greenschist-amphibolite facies fabric towards intrusive contacts. The dolerites are associated with syenites and granites of similar age, i.e. the ~1.22 Ga Vaggeryd syenite suite along the PZ (e.g. Johansson 1990) south of Lake Vättern (Fig. 1). Larsson and Söderlund (in review) dated the Småland-Taberg Fe-Ti mineralization at 1207 ± 6 Ma (Lu–Hf apatite age) showing that this magmatic event also involved emplacement of ultramafic, layered plutonic rocks.

The Blekinge-Dalarna dolerites (BDD) (c 098–095 Ga)

The BDD occupy a belt more than 700 km long and 150 km wide in the Sveconorwegian foreland region, extending from Blekinge in the south and disappearing beneath Caledonian cover rocks in the north (Fig. 1; cf. e.g. Gorbatschev et al. 1987). It is comparable in size to the CSDG but probably formed in a completely different tectonic setting. The swarm follows the eastern margin of the Sveconorwegian orogen on its eastern side, intruding Svecofennian and TIB rocks. This implies a relationship between dike intrusion and tectonic processes along the PZ-SFDZ and within the Sveconorwegian orogen to the west (see Wahlgren et al. 1994; Söderlund et al. 2004a). Petrographic and geochemical data of this swarm have been reported by e.g. Patchett et al. (1994), along wth Rb–Sr mineral dates of nine dolerite samples ranging between 850 and 1,050 Ma in age (Patchett 1978). Johansson and Johansson (1990) obtained Sm–Nd mineral dates in the same age range for dolerite dikes in Blekinge. Both Rb–Sr and Sm–Nd dates are associated with relatively large errors, about 20–50 Ma, so that the real age interval in which BDD intruded may be much narrower than implied by the means of Rb–Sr and Sm–Nd dates. The Karlshamn dolerite dike was recently dated at 954.1 ± 1.2 Ma by baddeleyite U–Pb chronology (Söderlund et al. 2004b).

The Göteborg dolerites (c 093 Ga)

The post-tectonic Göteborg dolerites were recently dated at 935 ± 3 Ma by U–Pb analysis of baddeleyite (Hellström et al. 2004). These dikes intruded preferentially along WNW trending brittle faults. The NW trending Dalsland dikes located about 150 km further to the north probably belong to the same swarm as well as norite-anorthosite complexes along the Swedish west coast (e.g. Scherstèn et al. 2000). A possible genetic relationship between the BDD and late-Sveconorwegian anorthosite and monzogabbro intrusions in Norway was discussed by Hellström et al. (2004).

Palaeozoic (c 280 Ma) basalts and dolerites in southern Sweden and Norway

About 300 million years ago, the western margin of Fennoscandia was subjected to extension believed to be connected with the Variscan orogenic activity in central Europe. Extensional E–W stress resulted in extensive magmatic activity and graben formation in the Oslo region, emplacement of dolerite sills in Västergötland (the Halle- and Hunneberg) and tectonic reactivation and intrusion of dolerite dikes along the NW trending Tornquist Zone in Skåne (Fig. 1). Typical for this generation of dolerites and volcanics are signs of crustal contamination e.g. the recognition of quartz-bearing olivine dolerites and high Sr- and low Nd–Pb-isotopic ratios in many intrusions (Neumann et al. 1988; Andersen and Knudsen 2000; Obst et al. 2004). K–Ar isotopic ages of Palaeo- and Mesozoic dolerites and basalts in Sweden are reported by Klingspor (1976).

Methods

U–Pb chronology

Samples for U–Pb geochronology were taken from the centre of dikes and sills, or from parts of intrusions coarser than elsewhere, e.g. pegmatitic patches. In conjunction with this sampling strategy we noticed a positive correlation between the size and abundance of baddeleyite grains and the sample grain-size, consistent with previous observations (Heaman and LeCheminant 1993). For instance, we noticed that fine-grained dolerites or samples from dikes and sills less than 10 m thick rarely contain baddeleyite. On the other hand, when baddeleyite was found, the least amount we extracted was approximately 150 grains. From this we conclude that a sample of about 1 kg either contains a sufficient number of baddeleyite grains for U–Pb TIMS analysis, or lacks baddeleyite completely. Approximately 2/3 of all the dolerite samples that we processed contained abundant baddeleyite.

Mineral separation was undertaken at the Department of Geology, GeoBiosphere Science Centre, Lund University, Sweden, employing the separation procedure of Söderlund and Johansson (2002). Approximately 150 to 500 baddeleyite grains/fragments were recovered from 0.5–1 kg samples. Usually it was possible to process two dolerite samples in one day, demonstrating the high efficiency of this separation technique. The baddeleyite grains are remarkably similar, dominated by perfectly transparent, dark brown and needle-shaped grains. In samples proximal to the Sveconorwegian orogen, i.e. the Värmland dolerite dikes and some of the 1.2 Ga Protogine Zone dolerites, a significant part of the baddeleyite population has secondary zircon overgrowths due to Sveconorwegian metamorphism. Alteration is seen mainly as a shift in colour from dark brown to paler brown (or sometimes yellow) and a decrease in transparency. The average dimension of the baddeleyite fragments is about 25 μm, though grains in some of the samples may reach as much as 50 μm in width. The grains selected for isotope analysis were carefully examined optically at high magnification so that only the best-quality grains were used for age determination. 20–40 baddeleyite grains in each fraction were combined for U–Pb ID-TIMS analysis.

U–Pb analytical protocol

Dissolution, ion exchange chromatography and mass spectrometry were performed at the University of Arizona. The baddeleyite grains were transferred to microcapsules and washed in warm 2 M HNO_3 and thereafter in warm H_20. Appropriate amounts of ^{205}Pb–^{233}U–^{235}U tracer and ultraclean HF and HNO_3 (in proportions 10:1) were added to the micro-capsules. The micro-capsules were placed in a teflon container that was fitted into a steel-jacketed high-pressure bomb. The baddeleyite grains were completely dissolved after 3 days at 210°C. The sample was dried, dissolved in 3.1 M HCl and loaded on 50 μl columns filled with anion exchange resin. Calibration of these columns demonstrates virtually perfect separation between Zr-Hf-Th-REE and U–Pb. The recovered Zr-Hf-Th-REE cut was saved for later determination of Hf isotope composition of the sample. Pb was loaded on a single Re filament with silica gel and phosphoric acid. Procedural blanks ranged from 2 to 5 pg Pb and <1 pg U. U and Pb isotopic measurements were performed in static and dynamic multicollector mode, respectively, using a Micromass Sector 54 solid source mass spectrometer. The ^{204}Pb was measured using a Daly electron multiplier with internal Daly-Faraday gain correction determined by peak switching of ^{205}Pb between the Daly and an off-axis Faraday collector. Mass fractionation is 0.06% per mass unit for Pb, determined by replicate analyses of NBS standards of SRM 981 and SRM 983. U fractionation was determined directly from the measured $^{233}U/^{235}U$ isotopic ratio. Initial Pb compositions were taken from the model of Stacey and Kramers (1975) for the age in question. Pb blank composition used was: $^{206}Pb/^{204}Pb = 19.10$, $^{207}Pb/^{204}Pb = 15.72$ and $^{208}Pb/^{204}Pb = 38.65$. The decay constants for ^{238}U (1.55125×10^{-11}) and ^{235}U (9.8485×10^{-10}) and the isotopic composition of U ($^{238}U/^{235}U = 137.88$) are those recommended by Steiger and Jäger (1977). Regression and calculation of uncertainties follow Ludwig (1980). The listed uncertainties in Pb/U ratios (Table 1) were calculated by propagating the within-run error for measured isotopic ratios, the uncertainty in fractionation (±0.05 for Pb and ±0.01 for U, absolute uncertainties), the uncertainty in Pb and U blank concentration ($\pm50\%$), and a ±0.05 absolute uncertainty in the Pb blank composition.

Lu–Hf and Sm–Nd isotopes

Initial Hf isotope compositions were determined for all dolerite samples dated in this study. For baddeleyite-bearing samples, and samples whose Nd isotope composition already was known from previous studies, the Hf isotope composition was determined from the same baddeleyite fraction as used for U–Pb analysis. For samples with no baddeleyite, Lu–Hf and Sm–Nd isotopic analysis was performed on whole-rock powders. These elements can easily be extracted in the first step of the Lu–Hf ion exchange, where Hf (plus matrix elements), and cuts rich in Lu and Sm–Nd are collected successively. The Lu–Hf analytical protocol for whole-rock samples essentially follows Patchett and Tatsumoto (1980) and is described in more detail by Söderlund et al. (2004b). Sm and Nd were measured on a VG-354 TIMS in dynamic mode (for details, see Patchett and Ruiz 1987). Sample preparation and Sm–Nd chemical separation followed methods previously reported by Vervoort and Patchett (1996). Analytical blanks were: Sm

Table 1 U–Pb data

Sample	Mineral type[a]	$^{208}Pb/^{206}Pb$	Pbc/Pbtot[b]	$^{206}Pb/^{204}Pb$ raw[c]	$^{207}Pb/^{235}U$ (corr)[d]	±2s% err	$^{206}Pb/^{238}U$	±2s % err	$^{207}Pb/^{235}U$ (age, Ma)	$^{207}Pb/^{238}U$	$^{206}Pb/^{207}Pb$
Breven-Hällefors dolerite dike											
Rejmyre	B1	0.010	0.0044	14676	3.7510	0.25	0.27643	0.24	1582.3	1573.3	1594.4
	B2	0.008	0.0036	18751	3.7843	0.14	0.27881	0.13	1589.4	1585.4	1594.8
	B3	0.011	0.0065	9927	3.6984	0.28	0.27239	0.27	1571.0	1552.9	1595.4
	B4	0.011	0.0047	13961	3.7572	0.21	0.27666	0.20	1583.7	1574.5	1595.8
Värmland dolerites											
Ölme	B1,y	0.020	0.0092	7345	3.5231	0.32	0.26549	0.32	1532.4	1517.9	1552.5
	B2	0.013	0.0335	1948	3.5901	0.43	0.26870	0.42	1547.3	1534.2	1565.4
	B3,y,op	0.022	0.0096	6887	3.3927	0.36	0.25490	0.35	1502.7	1463.7	1558.2
	B4,y,op	0.016	0.0127	5154	3.4208	0.21	0.25938	0.20	1509.2	1486.7	1540.9
Tuna dikes and age equivalents											
Bunkris	B1	0.021	0.0076	8933	3.0890	0.53	0.24435	0.49	1430.0	1409.3	1460.9
	B2	0.024	0.0028	24435	3.1264	0.36	0.24750	0.35	1439.2	1425.6	1459.4
	B3	0.022	0.0022	29955	3.1963	0.09	0.25275	0.07	1456.3	1452.6	1461.6
	B4	0.025	0.0044	15018	3.1984	0.20	0.25300	0.19	1456.8	1453.9	1461.0
Glysjön	B1	0.026	0.0033	20710	3.1566	0.34	0.24978	0.34	1446.6	1437.3	1460.3
	B2	0.018	0.0034	20349	3.1552	0.24	0.24966	0.24	1446.3	1436.7	1460.4
	B3	0.022	0.0045	14419	3.1920	0.35	0.25259	0.35	1455.2	1451.8	1460.2
	B4	0.028	0.0046	14027	3.2022	0.11	0.25325	0.10	1457.7	1455.2	1461.3
Gällsjön	B1	0.017	0.0059	11283	3.1817	0.20	0.25195	0.19	1452.7	1448.5	1458.9
	B2	0.012	0.0100	6575	3.2088	0.23	0.25373	0.22	1459.3	1457.7	1461.6
	B3	0.014	0.0159	4112	3.1886	0.41	0.25230	0.40	1454.4	1450.3	1460.4
	B4	0.014	0.0110	5857	3.0789	0.15	0.24631	0.14	1427.5	1419.4	1439.4
Central Swedish Dolerite Group (CSDG)											
Älvdalsåsen	B1	0.005	0.0139	4981	2.3419	0.40	0.20557	0.40	1225.1	1205.2	1260.3
	B2	0.005	0.0089	7662	2.4218	0.38	0.21227	0.37	1249.1	1240.9	1263.2
	B3	0.005	0.0096	7113	2.4680	0.24	0.21629	0.22	1262.7	1262.2	1263.5
	B4	0.006	0.0084	7835	2.4354	0.32	0.21343	0.31	1253.1	1247.1	1263.5
Ämån	B1	0.010	0.0099	6828	2.4885	0.15	0.21724	0.14	1268.7	1267.3	1271.1
	B2	0.021	0.0257	2491	2.4653	0.17	0.21510	0.15	1261.9	1255.9	1272.2
	B3	0.174	0.0071	8239	2.4603	0.16	0.21476	0.14	1260.4	1254.1	1271.3
	Z1	0.135	0.0208	3038	2.2378	0.67	0.19515	0.56	1193.0	1149.2	1273.1
	Z2	0.151	0.0063	9882	2.3064	0.58	0.20109	0.58	1214.3	1181.2	1273.6
Protogine Zone dolerites											
Bjärehalla	B1	0.024	0.0049	13752	2.2915	0.21	0.20565	0.20	1209.7	1205.6	1217.0
	B2,y,op	0.018	0.0075	9293	2.2452	0.48	0.20196	0.47	1195.3	1185.8	1212.4
	B3,y	0.026	0.0136	4915	2.2544	0.44	0.20292	0.44	1198.2	1191.0	1211.2
	B4,y,op	0.026	0.0221	3006	2.2520	0.60	0.20267	0.59	1197.4	1189.7	1211.4
Rumperöd	B1,y	0.024	0.0102	6710	2.2711	0.20	0.20450	0.19	1203.4	1199.4	1210.4
	B2,y	0.013	0.0124	5554	2.2886	0.30	0.20584	0.29	1208.8	1206.6	1212.6
	B3,y,op	0.079	0.0092	6680	2.2852	0.18	0.20557	0.16	1207.7	1205.2	1212.3
Blekinge-Dalarna dolerites (BDD)											
Bräkne-Hoby	B1	0.017	0.0032	21979	1.5324	0.11	0.15723	0.09	943.5	941.4	948.3
	B2	0.018	0.0138	5005	1.5383	0.22	0.15787	0.20	945.8	944.9	948.0
	B3	0.015	0.0043	15975	1.5198	0.32	0.15607	0.30	938.4	934.9	946.6
	B4	0.015	0.0058	11562	1.5500	0.47	0.15893	0.47	950.5	950.8	949.8
Falun	B1	0.010	0.0102	6921	1.5012	0.42	0.15433	0.41	930.9	925.2	944.4
	B2	0.012	0.0147	4693	1.4874	0.58	0.15289	0.55	925.2	917.1	944.6
	B3	0.015	0.0046	15047	1.4551	0.43	0.14954	0.43	912.0	898.4	945.0
	B4	0.018	0.0058	11906	1.5271	0.14	0.15689	0.13	941.4	939.4	945.8
Forserum	B1	0.011	0.0181	3733	1.6003	0.23	0.16242	0.20	970.3	970.2	970.6
	B2	0.012	0.0085	7902	1.5880	0.36	0.16110	0.34	965.5	962.9	971.5
	B3, y	0.009	0.0165	4126	1.6001	0.35	0.16253	0.34	970.2	970.8	969.0
	B4, y	0.012	0.0129	5239	1.6088	0.56	0.16327	0.48	973.6	974.9	970.7
Härsjön	B1	0.025	0.0296	2332	1.5275	0.73	0.15753	0.46	941.5	943.0	937.9
	B2	0.011	0.0128	5491	1.5443	0.51	0.15865	0.50	948.2	949.3	945.7
	B3	0.013	0.0153	4567	1.5536	0.39	0.15967	0.35	951.9	955.0	944.9
	B4	0.013	0.0100	6880	1.5257	0.45	0.15659	0.44	940.8	937.8	947.7
Lösen-Fäjö	B1	0.025	0.0099	6937	1.5292	0.17	0.15708	0.15	942.2	940.6	946.0
	B2	0.026	0.0366	1902	1.5437	0.56	0.15855	0.55	948.0	948.7	946.2
	B3	0.023	0.0243	2811	1.5495	0.58	0.15933	0.56	950.3	953.0	944.0

Table 1 (Contd.)

Sample	Mineral type[a]	$^{208}Pb/^{206}Pb$	Pbc/Pbtot[b]	$^{206}Pb/^{204}Pb$ raw[c]	$^{207}Pb/^{235}U$ (corr)[d]	±2s% err	$^{206}Pb/^{238}U$	±2s % err	$^{207}Pb/^{235}U$ (age, Ma)	$^{207}Pb/^{238}U$	$^{206}Pb/^{207}Pb$
Nilstorp	B1	0.011	0.0087	7839	1.5780	0.18	0.16078	0.14	961.6	961.1	962.6
	B2	0.013	0.0024	28817	1.5657	0.42	0.15927	0.41	956.7	952.7	965.9
	B3	0.012	0.0046	15019	1.5807	0.12	0.16085	0.11	962.6	961.5	965.2
	B4, y	0.011	0.0031	22491	1.5360	0.33	0.15627	0.32	944.9	936.0	965.6
Nornäs	B1	0.025	0.0321	2168	1.5299	0.62	0.15721	0.59	942.5	941.2	945.3
	B2	0.016	0.0082	8375	1.5383	0.19	0.15789	0.17	945.8	945.0	947.7
	B3	0.017	0.0087	7928	1.5258	0.54	0.15674	0.53	940.8	938.6	946.0
Särna	B1	0.026	0.0187	2312	1.6451	0.32	0.16628	0.31	987.7	991.6	979.0
	B2	0.024	0.0111	4144	1.6498	0.20	0.16701	0.11	989.5	995.6	975.9
	B3	0.052	0.0569	964	1.6523	0.65	0.16691	0.62	990.4	995.1	980.2
	B4	0.017	0.0012	4278	1.5482	0.34	0.15659	0.32	949.8	937.8	977.6

[a]All baddeleyite grains are clear and dark brown unless specified otherwise. *y* Yellow, *op* opaque. Grains/fragments (20–40) were analysed in each fraction
[b]*Pbc* common Pb; *Pbtot* total Pb (radiogenic + blank + initial)
[c]Measured ratio, corrected for fractionation and spike
[d]Isotopic ratios corrected for fractionation (0.06% per amu for Pb), spike contribution, blank (2–5 pg Pb and < 1 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at the age of the sample

< 25 and Nd < 80 pg, less than 0.1% for these elements in all samples.

Sm–Nd analysis of the Rotsjö, Skutskär, Åstön and Draget samples were conducted at the Laboratory of Isotope Geology (LIG) at the Swedish Museum of Natural History in Stockholm. Measurements were done on a Finnigan MAT 261, and Nd ratios were normalized to $^{146}Nd/^{144}Nd = 0.7219$. Estimated errors for $^{147}Sm/^{144}Nd$ are < 0.5%. Due to a considerable time difference and variations in the La Jolla measurements over time for the individual samples measured at LIG, a normalization of the measured values to a $^{143}Nd/^{144}Nd$ La Jolla value of 0.511854 was applied in order to allow a correct comparison. During measurement of the two most recently analysed samples (Rotsjö and Skutskär) the average measured $^{143}Nd/^{144}Nd$ ratio of the La Jolla Nd-standard was 0.511842 ± 8 ($n = 7$; 2σ). An extensive description of the analytical details is given in Andersson (1997a) and Andersson et al. (2002).

Lu–Hf analytical protocol of baddeleyite samples

Because baddeleyite has very high Hf contents (10,000–15,000 ppm) compared to Lu (usually less than some tens of ppm), the effect on initial $^{176}Hf/^{177}Hf$ from the contribution of in situ production of ^{176}Hf by decay of ^{176}Lu is usually very minor. $^{176}Lu/^{177}Hf$ in baddeleyite from three different dolerite samples reported previously (Schärer et al. 1997; Scherer et al; 2001; Söderlund et al. 2004b) is equal to 0.0001, which corresponds to less than 0.1 ϵ_{Hf} unit between corrected and uncorrected isotopic ratios for a 1.6 Ga old sample. For younger samples the affect is even less. Baddeleyite in one sample, the Sorkka dolerite reported by Söderlund et al. (2004b), is much richer in Lu than other samples ($^{176}Lu/^{177}Hf = 0.00066$). Such high ratios would bias the initial ϵ_{Hf} value by as much as 0.7 ϵ_{Hf} units in a 1.6 Ga old sample. Providing that the baddeleyite grains in the Sorkka dolerite are anomously Lu-rich, the present-day $^{176}Hf/^{177}Hf$ of baddeleyite can generally be assumed to equal its initial $^{176}Hf/^{177}Hf$, and neither isotopic dilution nor separation of Lu from Hf is then necessary. Only a single clean-up step is required for reducing mass interference on mass 176 by natural Yb and Lu.

The collected Zr-Hf-REE wash from the HCl chemistry for U–Pb was dried and dissolved in 1 ml 1.0 M HCl/0.1 M HF. Hf was purified from REE on a column with 1.5 cm^3 of cation resin AG50W-X12. Hf is eluted by 2 resin volumes of 1.0 M HCl/0.1 M HF and evaporated to dryness. One drop of $HClO_4$ was added for removing organic matter that may remain after the ion exchange. Total procedural blank was 10 pg Hf that corresponds to < 0.1% of all Hf in a sample. This means that samples do not need blank corrections.

Hf isotopic analysis was conducted in static mode on a Micromass Isoprobe multiple-collector inductively coupled plasma mass spectrometer (MC-ICPMS). For both whole-rock and baddeleyite samples, we used an ARIDUS desolvating nebulizer for increasing yield. Samples were dissolved in 1 ml 2% HNO_3 before mass spectrometric analysis, and this amount of solution yielded around 60 Hf isotope ratios. Duplicate runs of JMC-475 ($n = 17$) gave an error in $^{176}Hf/^{177}Hf$ of ± 0.000025, defined by two standard deviations of the population. This is about the same ($= 0.000020$) as the variation in $^{176}Hf/^{177}Hf$ observed in duplicate runs of 11 dolerite samples, performed both on baddeleyite grains and whole rock powders (Table 3). This difference is about twice as much as the within run precision at 2 standard deviations of the mean of individual analysis (approximately ± 0.000010) and demonstrates excellent reproducibility for our Hf isotope analyses.

Possible isobaric interferences on ^{176}Hf were evaluated by monitoring ^{175}Lu and ^{172}Yb. The amounts of Yb in the Hf runs were very small and only minor corrections for Yb interference was necessary

($^{176}Yb/^{177}Hf$ was typically <0.000020). No corrections for Lu interferences were necessary. The $^{176}Hf/^{177}Hf$ ratio was normalized for mass fractionation to $^{179}Hf/^{177}Hf = 0.7325$ using an exponential correction. Cup efficiencies were set to unity before sample analysis. A 30 ppb solution of JMC-475 was analyzed until no significant changes in Hf isotopic ratios between blocks were observed. The cup efficiencies were thereafter adjusted to match the recommended values of the JMC-475 standard solution; $^{176}Hf/^{177}Hf = 0.282160$, $^{178}Hf/^{177}Hf = 1.467170$, $^{179}Hf/^{177}Hf = 0.7325$ and $^{180}Hf/^{177}Hf = 1.88666$ (Vervoort et al. 2004).

Long-term observations of duplicate Sm–Nd analyses and frequent runs of La Jolla standard lead us to assign uncertainties of 0.2% and ± 0.000015 (2σ) to $^{147}Sm/^{144}Nd$ and $^{143}Nd/^{144}Nd$, respectively. These numbers exceed the within-run errors (two standard errors of the mean) by a factor of approximately 2. Duplicate analyses of two samples, the Åker and Romeleklint, deviate by 0.4 and 0.2 ϵ_{Nd} units. The dolerites are assigned total uncertainties of about 0.5 epsilon units for both Nd and Hf.

U–Pb Results

The location of dolerite dikes and sills sampled for U–Pb chronology are shown in Fig. 1. Table 2 summarises the U–Pb age results and gives the exact locations of sample sites given in coordinates of the Swedish National Grid. The U–Pb results are shown graphically on Concordia diagrams in Fig. 2. All stated U–Pb dates are interpreted to date the magmatic crystallization of the rocks. The U–Pb data for the majority of samples are concordant to moderately discordant. The data for a few samples, however, scatter beyond analytical uncertainties (i.e. MSWD $=2$). Excess scatter occurs mainly in samples where secondary alteration has been observed. The discordance seen in some BDD samples is entirely attributed to modern Pb-loss as indicated by lower intercepts (within error of 0 Ma) and by the absence of any major post-Sveconorwegian high-T event in southern Sweden (e.g. Å berg 1978). We employed a forced lower intercept of 0 ± 400 Ma in the regression of the BDD samples. Older samples located within or close to the SFDZ/PZ could potentially have undergone earlier isotopic disturbances. This is, for instance, indicated by local disturbance of the K–Ar isotope system for some mafic rocks in Dalarna (Priem et al. 1968), and remagnetization in mid-Proterozoic dolerites in southern Sweden (Bylund 1992). Therefore, regression of pre-Sveconorwegian dolerite samples close to the Sveconorwegian boundary zones was performed with unforced lower intercepts, unless stated otherwise. Because baddeleyite samples were not weighed no U, Th or Pb concentrations are given. The $^{208}Pb/^{206}Pb$ ratios are lower than 0.026 for 57 for the 59 baddeleyite fractions that we analysed. These numbers basically reflect that baddeleyite takes up Th much less readily than does zircon.

Breven-Hällefors dolerite dikes

The baddeleyite grains recovered from the Rejmyre dolerite dike are dark brown, transparent and with no signs of alteration. Despite the seemingly good quality, all four fractions are discordant by 0.6 to 2.7% (calculated as the discordance of mean $^{207}Pb/^{206}Pb$ and $^{206}Pb/^{238}U$ dates). The upper intercept of $1{,}595 \pm 3.5$ Ma, interpreted as the age of the Rejmyre dolerite, is similar to the baddeleyite U–Pb result of the Breven dolerite ($1{,}602 \pm 2$ Ma; J. Connelly, personal communication).

Värmland dolerites and age equivalents in south-central Sweden

The flat-lying Ölme dolerite follows an irregular pattern over an approximately 5×12 km area immediately to the northeast of Lake Vänern (Fig. 1). Many of the baddeleyite grains in the Ölme sample have thin rims of secondary zircon. The presence of small garnets in the sample further documents the imprint of metamorphism. Baddeleyite fractions show considerable discordance and $^{207}Pb/^{206}Pb$ dates vary between 1,564 and 1,541 Ma, the former comprising analysis of the best grains with either no or very little zircon overgrowth. Regression using the oldest $^{207}Pb/^{206}Pb$ dates of two baddeleyite fractions gives an upper intercept of $1{,}569 \pm 3$ Ma and a lower intercept close to 0 Ma. We consider this result as the best age estimate of the Ölme intrusion. Similar scatter in baddeleyite data was documented for two dolerite dikes only a few km south of the Ölme sample (Wahlgren et al. 1996). Their age of ~1,568 Ma is, however, identical with our result of the Ölme dolerite and suggests this to be the true age of the Värmland dolerite dikes. Besides the 1,572–1,562 Ma meta-dolerites further to the south along the PZ, other age equivalents are dolerites and quartz porphyry dikes (1,577–1,540 Ma) and granites (1,562–1,590 Ma) in SW Finland (Suominen 1991).

Tuna dolerites and age equivalents

Three dikes sampled in northern Dalarna, i.e. the Bunkris, Glysjön and the Gällsjön samples, give identical dates of 1461.7 ± 1.0, 1461.2 ± 1.2 and 1461.2 ± 1.4 Ma (Fig. 2). According to mapping (Hjelmqvist 1966) the dolerite samples taken at Bunkris and Glysjön are from one and the same NW-trending dike reaching at least 20 km in length. Further south in Dalarna, Lundström et al. (2002) reported a U–Pb date of $1{,}474 \pm 4$ Ma for a porphyry sample belonging to the Gustafs porphyries, hence very similar to the baddeleyite U–Pb dates reported for the dolerites in the north. These results imply a genetic kinship between the Tuna dikes and dolerite dikes dated in this study. The here dated 1,462–1,461 Ma intrusions manifest an important tectonic link to coeval mafic intrusions further west in the

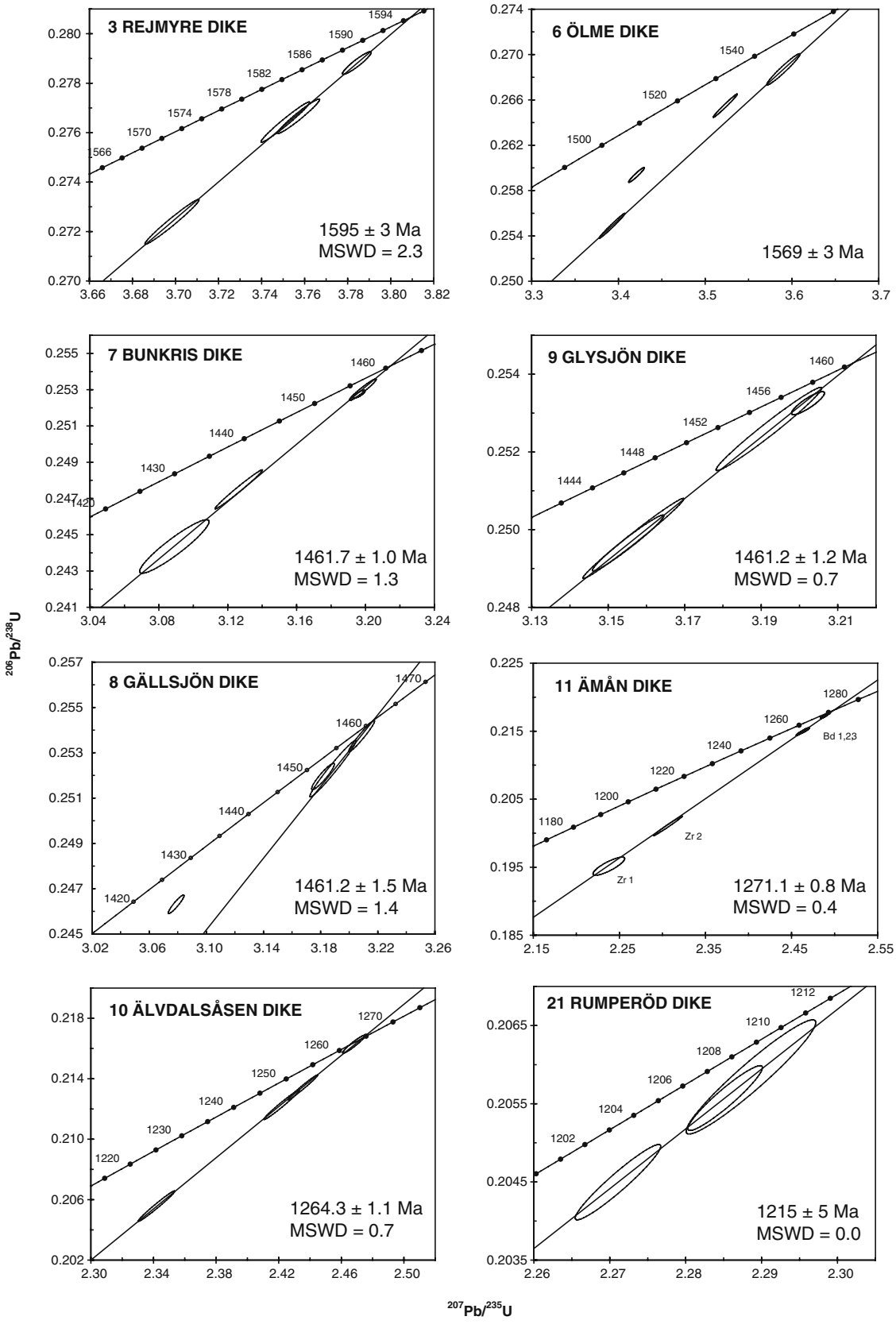


Fig. 2 Concordia plots of U–Pb data from baddeleyite and a few zircon fractions in dolerite dikes and sills from 6 different generations in central Fennoscandian Shield, Sweden. *Number* before sample name refers to sample location given in Fig. 1. *Ellipses* indicate error at the 95% confidence level for the analyses

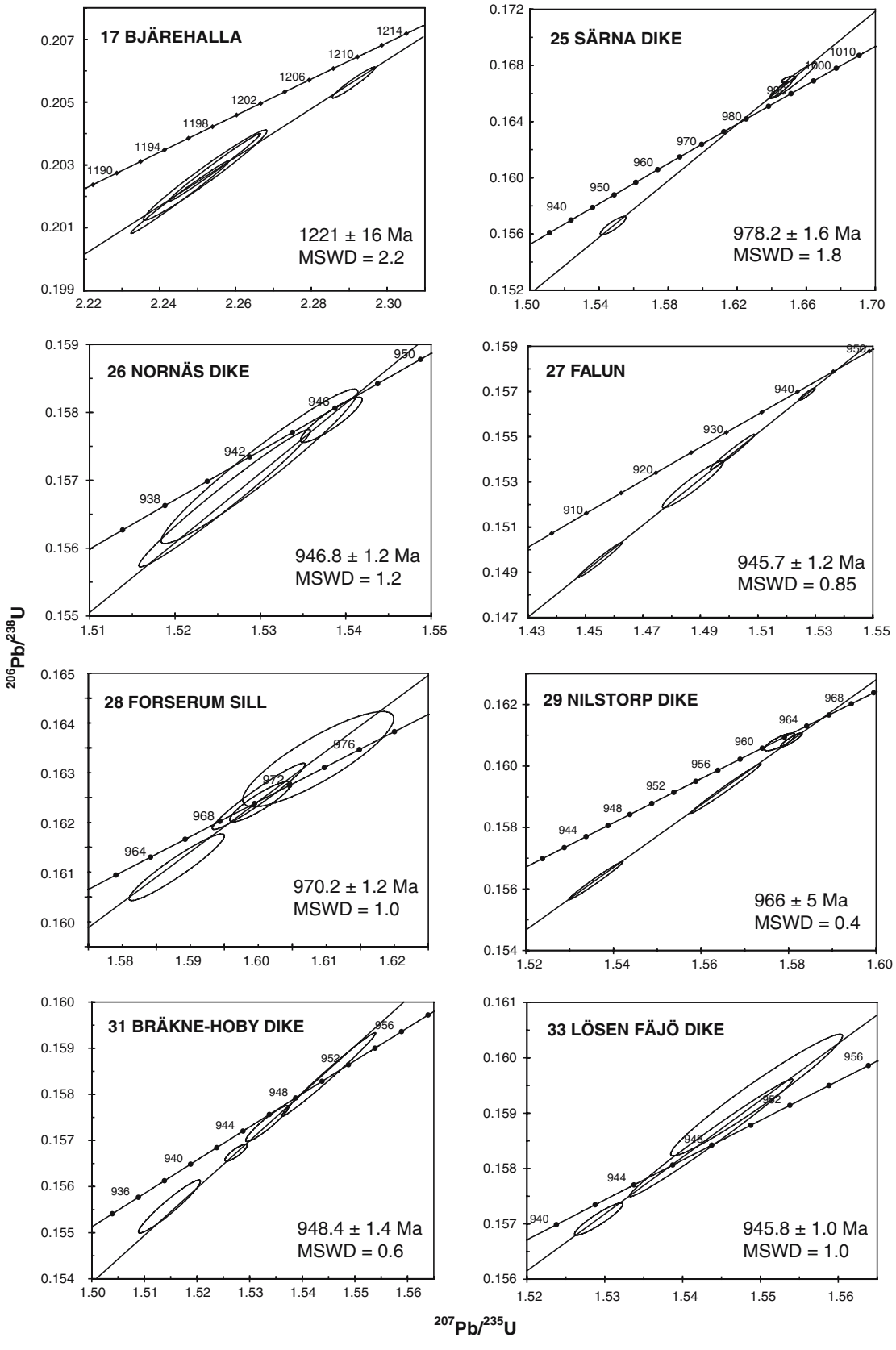

17 BJÄREHALLA
1221 ± 16 Ma
MSWD = 2.2
25 SÄRNA DIKE
978.2 ± 1.6 Ma
MSWD = 1.8
26 NORNÄS DIKE
946.8 ± 1.2 Ma
MSWD = 1.2
27 FALUN
945.7 ± 1.2 Ma
MSWD = 0.85
28 FORSERUM SILL
970.2 ± 1.2 Ma
MSWD = 1.0
29 NILSTORP DIKE
966 ± 5 Ma
MSWD = 0.4
31 BRÄKNE-HOBY DIKE
948.4 ± 1.4 Ma
MSWD = 0.6
33 LÖSEN FÄJÖ DIKE
945.8 ± 1.0 Ma
MSWD = 1.0
206Pb/238U
207Pb/235U

Fig. 2 (Contd.)

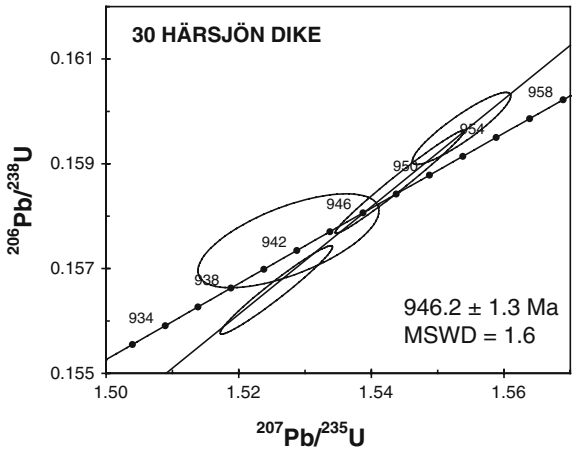


Fig. 2 (Contd.)

Sveconorwegian orogen such as the 1,457 ± 6 Ma Orust and Kattsund-Koster dikes (cf. Åhäll and Connelly 1998).

Central Scandinavian Dolerite Group

The Älvdalsåsen and Ämån dolerites are observed in the same area as the 1,461 Ma dolerites. They give precise U–Pb dates of 1264.3 ± 1.1 and 1271.1 ± 0.8 Ma (Fig. 2). The three baddeleyite and two zircon fractions analysed in the Ämån sample are aligned along a common discordia line with an unforced lower intercept close to the origin. Zircon in this sample records greater modern Pb-loss than baddeleyite. In accordance with palaeomagnetic data indicating small time differences between different CSDG complexes in Scandinavia (Elming and Mattsson 2001), we obtain an age gap of at least 13 million years between dolerites in Dalarna and Satakunta (south-western Finland; Suominen 1991). CSDG is coeval with the Gardar and Mackenzie dikes in Greenland and North America, respectively. The occurrences of tholeiitic to alkaline c. 1.27–1.25 Ga dolerite complexes in Greenland and Scandinavia reflect a tensional regime, which Elming and Mattsson (2001) interpreted as part of the beginning stage of break up between Laurentia and Fennoscandia.

Protogine Zone dolerites

Dolerites were sampled at Röshult, Rumperöd, Bjärehalla, Steninge-Boda and Bjärlöv (Fig. 1). Only the Rumperöd and Bjärehalla samples contained abundant baddeleyite grains, and were dated at 1,215 ± 5 and 1,221 ± 16 Ma (Fig. 2). Considering that most baddeleyite grains show signs of alteration it is not surprising that none of the fractions are concordant. Similar dates (between 1,222 and 1,219 Ma) were obtained by Ask (1996) on five samples of the Vaggeryd syenite and associated rocks, located some tens of km south of Lake Vättern along the PZ. These results strengthen previous inferences (see Larsson and Söderlund in review) that the main pulse of syenitic, granitic and mafic magmatism along the PZ took place within a time period of approximately 5 million years.

The Blekinge-Dalarna dolerites (BDD)

Sampled dikes comprise the Särna, Nornäs and Falun dolerites in Dalarna, the Forserum and Nilstorp dolerites in Småland, and the Bräkne-Hoby, Lösen-Fäjö and Härsjön dikes in Blekinge (Fig. 1). Notably is that the dike at Särna is not a member of the "Särna-type diabases" described by Hjelmqvist (1966), which are about 1.26 Ga in age. All samples except the Nilstorp dolerite give robust U–Pb dates with 2σ errors below ± 1.5 Ma (Fig. 2). The MSWDs are about 1.5 or less for these samples, suggesting that the observed scatter can be assigned to analytical uncertainties alone (cf. Wendt and Carl 1991). The Nornäs and Falun dikes in Dalarna give precise and identical U–Pb dates of 946.8 ± 1.2 and 945.7 ± 1.2 Ma whereas the Särna dike is dated at 978.2 ± 1.6 Ma, the oldest BDD dike dated in this study. The Nornäs analyses are basically concordant, whereas the baddeleyite analyses from the Falun dolerite are variably discordant (1–5%). Three of four fractions of the Särna dike plot slightly above the Concordia (Fig. 2). Ages of 966 ± 5 and 970.2 ± 1.2 Ma are obtained for the Nilstorp and Forserum sills in Småland. A significant fraction of baddeleyite grains in these two samples show signs of alteration, and the U–Pb data of the Nilstorp dolerite sample are more discordant than the U–Pb results of other BDD samples. Though grains with obvious signs of alteration were avoided during hand picking, some may have contained invisible thin rims of secondary zircon, or were isotopically disturbed. Including all four fractions in the regression gives an upper intercept at 965 ± 5 Ma (MSWD = 2.5). One data point of the Nilstorp sample falls slightly left of the regression line. An almost identical result (966 ± 5 Ma) is attained if this fraction is excluded, but the MSWD value is reduced to 0.43. We prefer the latter result, as the age of the Nilstorp sill. An even older age, at 970.2 ± 1.2 Ma, is attained for the Forserum sill located ~20 km west of the Nilstorp sample. The three dolerites sampled in Blekinge resemble the Nornäs and Falun dolerites in Dalarna by having identical U–Pb dates: 948.4 ± 1.4, 945.8 ± 1.0 and 946.2 ± 1.3 Ma for the Bräkne-Hoby, Lösen-Fäjö and the Härsjön dikes, respectively. Only moderate discordance (between −1 and +1%) is observed for these samples.

Discussion

Comparison between U–Pb, Rb–Sr and Sm–Nd ages

Figure 3 summarizes the U–Pb results of this study and earlier reported Rb–Sr and Sm–Nd isochron dates of

dolerites in Scandinavia with references given in the caption text. Only age results considered by these workers to be geologically significant are included. Most Rb–Sr dates are from Patchett (1978) and involve isotope analysis that combines a whole rock with a biotite fraction, or alternatively, by combining different minerals in a sample. The exception is the age for the Tuna dikes (1,388 ± 50 Ma), which is constrained from whole rock samples of dikes with varying Rb/Sr ratios. Four Rb–Sr mineral and all the Sm–Nd ages are from Johansson and Johansson (1990).

Some general inferences can be drawn from Fig. 3. First, uncertainties of individual ages are usually more than an order of magnitude higher for Rb–Sr and Sm–Nd compared to U–Pb. Second, despite using a lower value of the decay constant, which increases Rb–Sr dates by 1.27% relative to the currently used value of λ^{87}Rb, Rb–Sr ages of dolerites belonging to the older suites are younger than U–Pb dates. The value of λ^{87}Rb we used for recalculating the Rb–Sr dates is $1.402 \pm 0.008 \times 10^{-11}$ years^{-1} (Minster et al. 1982), which is about 1.3% lower than the currently used value (1.42×10^{-11} years^{-1}; Neumann and Huster 1974). The lower value of λ^{87}Rb represents the best agreement between the whole-rock Rb–Sr data of chondrites and their U–Pb ages of 4.555 ± 0.010 Ga. Owing to the fact that error in decay constants determined by calibration experiments causes a systematic shift by the same magnitude and in the same direction, it is more logical to use the value of 1.402×10^{-11} years^{-1} when, as here, Rb–Sr and U–Pb dates are compared (cf. Amelin and Zaitsev 2002). The Sm–Nd method yields, without exception, much lower ages than the U–Pb method.

If we consider Rb–Sr whole rock-biotite ages only, both the Breven-Hällefors suite and the CSDG constitute two coherent data sets with ages in the 1,529–1,565 Ma (3 samples) and the 1,228–1,263 Ma (6 samples) intervals, respectively. The means are about 50 and 25 Ma younger than the U–Pb ages of these two dolerite suites. This discrepancy could potentially relate to slow cooling in combination with relatively low effective closure temperatures or to post-magmatic isotopic disturbance. However, we consider that none of these mechanisms is very plausible. First, biotite K–Ar and Ar–Ar studies in areas intruded by the Breven-Hällefors suite and CSDG give ages of 1.67–1.70 Ga (Page et al. 2004), which coincide with the age of large igneous batholiths in the Dala and Rätan igneous provinces (cf. Högdahl et al. 2004). This largely precludes regional post-1.67 Ga re-heating above 300°C. Furthermore, if the Jotnian sedimentation in Dalarna was accomplished at c. 1.46 Ga (cf. Nyström, 2004), the present peneplain in Sweden and Finland must have been very close to the present surface already at that time. Accepting shallow crystallisation results in an impossibly slow cooling rate (18 and 45°C/Ma) of mafic magmas from ~1,200 to 300°C (the approximate closure temperature of the Rb–Sr biotite system) to account for the U–Pb and Rb–Sr age discrepancy of the two dolerite suites. Slow cooling cannot be a major control of the observed age discrepancy.

A common characteristic of biotite in Proterozoic dolerites in Sweden is a slight alteration to chlorite. Isotopic disturbance related to low-temperature fluid activity would have a similar affect on the Rb–Sr isotopic system in potassium feldspar, and Sm–Nd in

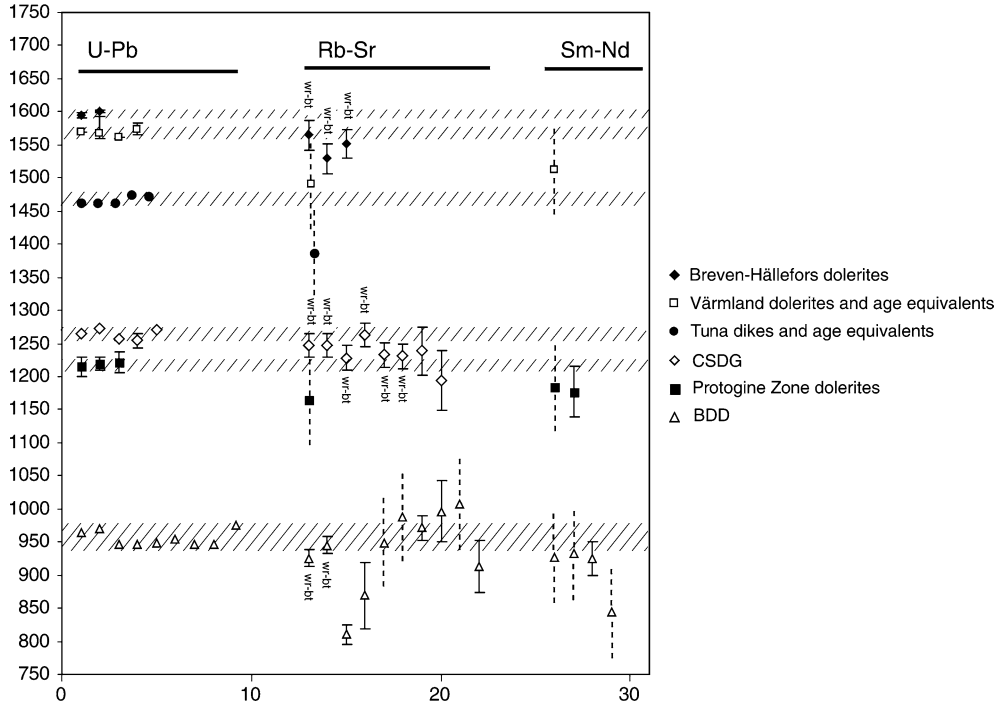


Fig. 3 Summary of U–Pb, Rb–Sr and Sm–Nd dates of Meso- and Neoproterozoic dolerite dikes and sills in Scandinavia. The *broad horizontal lines* mark "true" ages of the rocks, considered to correspond to U–Pb dates. *Error bars* are defined by two standard deviations. Uncertainties exceeding ± 50 Ma are marked with *dashed lines*. *Symbols* that lack error bars have uncertainties less than ± 5 Ma. U–Pb data are from J. Connelly (personal communication), Wahlgren et al. (1996), Söderlund et al. (2004), C. Möller (personal communication), Lundström et al. (2002), Suominen (1991), Ask (1996), Claesson and Kresten (1997) and this study. Rb–Sr and Sm-Nd data are from Patchett (1978) and Johansson and Johansson (1990). Rb–Sr ages have been recalculated using $\lambda^{87}\text{Rb} = 1.402 \pm 0.008$ (Minster et al. 1982)

pyroxene, i.e. by decreasing the slope (age) of the isochron. From U–Pb data we know that dolerite magma intruded repeatedly at c. 1.60, 1.56, 1.46, 1.26–1.27, 1.22 and 0.98–0.95 Ga, and any of these events could have caused fluid activity in crustal rocks by a moderate increase of temperature. This scenario would be compatible with a larger age difference observed between Rb–Sr and U–Pb dates for older dolerites compared to younger: 50 Ma for the Breven-Hällefors, 25 Ma for CSDG, and virtually 0 Ma difference for the BDD.

Mantle sources and processes

Figure 4 shows initial Nd and Hf ratios expressed as epsilon values versus age (1.6 to 0.28 Ga) for the ten dolerite suites from the southern and central parts of the Fennoscandian Shield (Table. 3 and 4). In order to expand the Hf and Nd database of mantle-derived Proterozoic rocks we also carried out isotopic analyses of dolerites not dated in this study, but which can be assigned to a certain dolerite generation based on previous chronological studies or other criteria. We also present Hf isotope data for six Palaeozoic samples including the Hunneberg basalt and NW trending dolerites in Skåne. References to additional Nd data are given in the figure caption.

Figure 5 shows that our data lie in the centre of the terrestrial array determined from the Hf and Nd isotopic compositions of various types of crustal rocks (Vervoort et al. 1999). The total maximum range in ϵ_{Nd}or ϵ_{Hf} for each Proterozoic dolerite generation observed here is 3–4 units, which is slightly higher than that normally observed in pristine dolerite samples (spread of 2–3 U) with no crustal contamination (e.g. Patchett et al. 1994). Our Hf and Nd data are in good agreement with the data of earlier isotopic studies of the same swarms of Proterozoic dolerites in Fennoscandia (e.g. Claesson 1987; Johansson and Johansson 1990; Rämö 1990; Patchett et al. 1994; Andersson 1997a). There are no clear differences in initial Hf or Nd isotope compositions between BDD samples from different areas. Similarly, the CSDG shows only limited regional variation. However, Andersson (1997a) obtained a somewhat lower $\epsilon_{Nd}(1.27)$ of −0.7 for a sample from the basalt layer intercalated in the Jotnian sandstone in the Gävle graben. Though this sample may be more contaminated by crustal material, it is also possible that it is older, c. 1.46 Ga. Nyström (2004) reported relatively low $\epsilon_{Nd}(1.46)$ values for a number of samples of Jotnian basalts from south central Sweden in the range −2 to 0. The Palaeozoic samples, on the other hand, have strongly variable initial compositions with as much as 17 and 12 ϵ_{Hf} and ϵ_{Nd} units, respectively. Such large variations agree with compositional and isotopic evidence demonstrating significant crustal contamination for some dolerites and basalts of this age in Scandinavia (Neumann et al. 1988; Andersen and Knudsen 2000; Obst et al. 2004).

Hf data for all Proterozoic samples investigated here yield slightly depleted to chondritic signatures. The same is the case for the Nd data, which however also extend to more enriched (negative ϵ_{Nd}) signatures for some dolerite suites. Some systematic differences between the different generations can be discerned. For example, the CSDG has higher ϵ_{Nd} and ϵ_{Hf} than both the older and younger dolerites further south, e.g. the c. 1.3 Ga Västergötland and the c. 1.22 Ga Protogine Zone dolerites, respectively. Further, the BDD center on chondritic ϵ_{Nd} values while the 935 Ma Göteborg dolerites indicate higher values (+3.7 for the Heden dolerite). These regional variations reflect either variations in the subcontinental mantle or in the amount of contamination during intrusion or metamorphism.

Initial ϵ_{Nd} isotopic compositions for Subjotnian rapakivi-related dolerites and associated mafic plutonic rocks vary strongly across the shield (Fig. 5). The c. 1.65 Ga dolerite swarms in SE Finland show essentially chondritic compositions: 0 ± 1.5 (Rämö 1990, 1991), the c. 1.58 Ga dykes in SW Finland (including the Nordingrå anorthosite complex) tend to be more negative: −3 to 0 (Rämö 1991; Fröjdö et al. 1996; Lindh et al. 2001), while the 1.53–1.50 Ga Subjotnian dykes in central Sweden are strongly negative: −10 to −4.5 (e.g. Andersson 1997b; Andersson et al 2002). It is important to note that the two latter groups occur in the same geographical area as the CSDG. Andersson et al. (2002) attributed the strongly enriched character of the central Swedish 1.53–1.50 Ga dykes to contamination of 'non-enriched' mantle melts by pyroxene + plagioclase-rich depleted Archaean lower crustal material, but did not exclude Archaean-enriched mantle. Recent work in the Jotnian Lake Ladoga basin in the eastern part of the Shield, close to the Archaean border, have shown that the dolerites and intercalated basalts are c. 1.46 Ga in age rather than c. 1.26 Ga, as previously assumed, and have strongly negative initial ϵ_{Nd} values: −9.5 to −4, which have tentatively been related to a partly Archaean-enriched mantle source (Rämö et al. 2004). Initial ϵ_{Hf} values (c. +4.5) determined here for 1.46 Ga dikes in Dalarna translate into ϵ_{Nd} compositions of c. +1 (calculated from the relation $\epsilon_{Hf} = 1.34\epsilon_{Nd} + 2.82$, cf. Vervoort et al. 1999), and hence very different from coeval mafic rocks in the Lake Ladoga basin.

The juvenile 1.95–1.7 Ga Svecofennian metaigneous crust is the main available crustal contaminant in the areas where the investigated dolerites occur. The upper limit of its ϵ_{Nd}-evolutionary array intersects CHUR at c. 1.6 Ga following a relatively steep trend to about −6 at 0.9 Ga (Fig. 4; see compilation in Andersson et al. 2002). Svecofennian metasedimentary rocks are in general more enriched in older components than the metaigneous crust. This is corroborated by the presence of ~30% Archaean detrital zircons in these rocks, while the majority of grains fall in the range 2.1–1.86 Ga (e.g. Huhma 1987; Claesson et al. 1993; Lahtinen et al. 2002). Except for central Sweden, Archaean components have

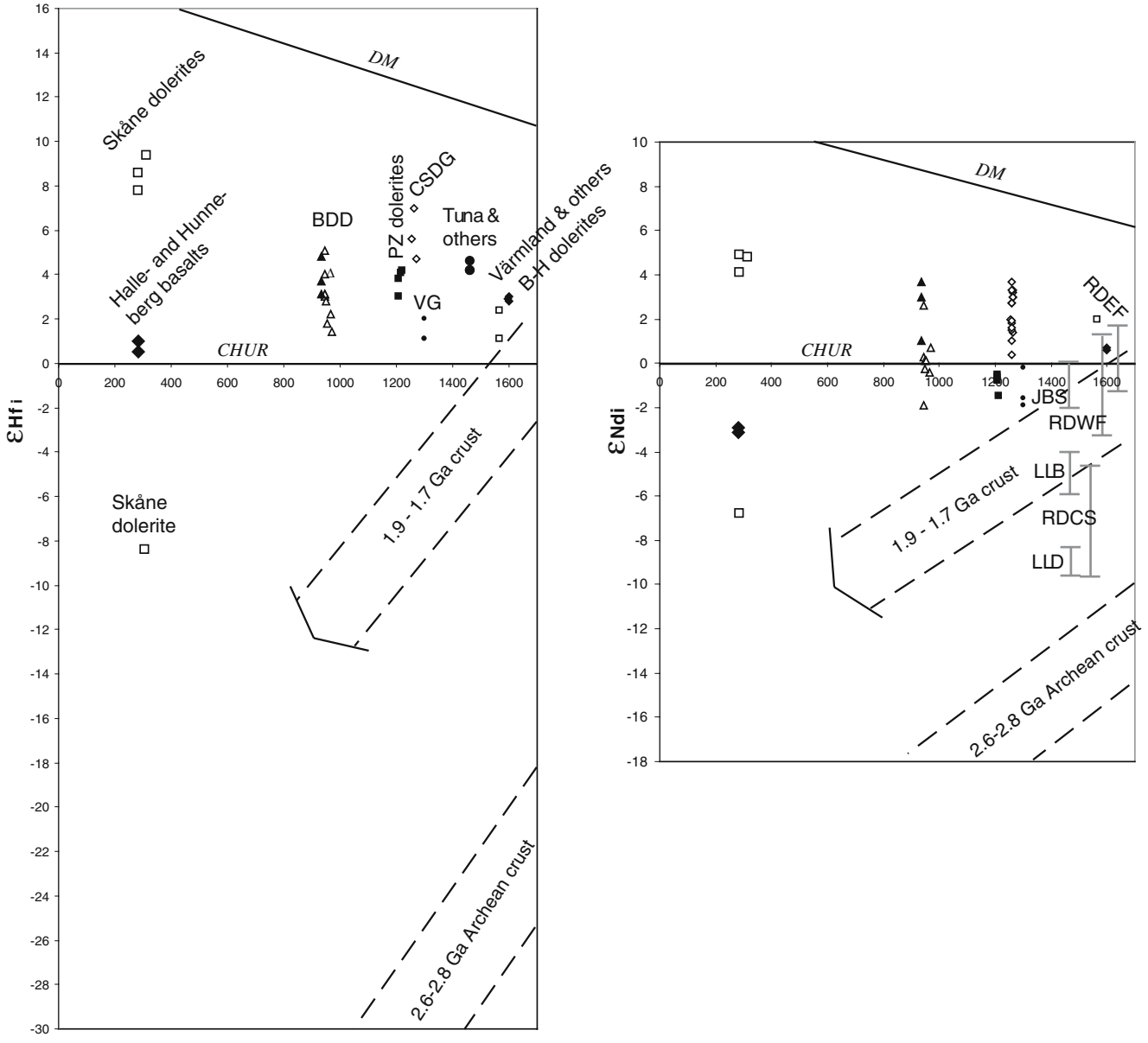


Fig. 4 Summary diagrams of ϵNd and ϵHf versus time of samples investigated in this study. Names of dolerites have been put above *symbols in left diagram* except the Göteborg dolerites which are represented by *dark triangles* under BDD. *VG* Västergötland dolerites, *B-H* Breven-Hällefors dolerites. *Bars in the right diagram* depict initial Nd isotope composition for *JBS* Jotnian basalts, south-central Sweden (Nyström 2004), *RDEF* Rapakivi-dolerites SE Finland (Rämö 1990; 1991), *RDWF* Rapakivi-dolerites SW Finland (Rämö 1990; Fröjdö et al. 1996; Lindh et al. 2001), *RDCS* Rapakivi-dolerites central Sweden (Persson 1999; Andersson et al. 2002), *LLB* Lake Ladoga basalts (Rämö et al. 2004) and *LLD* Lake Ladoga dolerites (Rämö et al. 2004). Additional Nd from: Claesson (1987), Patchett et al. (1994), Johansson and Johansson (1990), Larsson and Söderlund (in review), and F. Hellström (Göteborg, personal communication). Initial ϵ_{Nd} values reported by Claesson (1987), and Johansson and Johansson (1990) have been adjusted according to new U–Pb dates (this study). *Lines* marked DM and CHUR signify depleted mantle and chondritic uniform reservoir. Nd depleted mantle curve from DePaolo (1981). Hf and Nd depleted *mantle curves* are based on present day ϵ_{Hf} and ϵ_{Nd} values of 18 and 12, respectively, and 6.5 and 4 at 2.7 Ga (Vervoort and Blichert-Toft 1999). The Hf and Nd *evolution lines* are shown for 1.9–1.7 Ga and 2.6–2.8 Ga crustal rocks (data summarized in Andersson et al. 2002). CHUR reference values are given in the footnotes of Table. 3 and 4

not been identified in the Svecofennian metaigneous crust (Andersson et al. 2002). Potentially, the variation in initial ϵ_{Hf} and ϵ_{Nd} values within and between the presently investigated dolerite suites can be accounted for by variable degrees of contamination by these available crustal components. However, based on the characteristic and relatively limited range in Nd and Hf isotopic compositions for each individual suite we preclude very local mixing between depleted and strongly enriched crustal sources for this variation. Mixing of rocks with large differences in Nd and Hf isotope compositions is expected to cause extensive scatter in mag-

Table 2 Summary of U-Pb ages and sample locations

Sample	Age ± 2σ (Ma)	Map reference (Swedish National Grid)
Blekinge-Dalarna dolerites (BDD)		
Falun	945.7 ± 1.2	67196/14876
Lösen–Fäjö	945.8 ± 1.0	62278/14923
Härsjön	946.2 ± 1.3	62352/14600
Nornäs	946.8 ± 1.2	68147/13617
Bräkne–Hoby	948.8 ± 1.4	62371/14609
Nilstorp	966 ± 5	63910/14426
Forserum	970.2 ± 1.2	63994/14221
Särna	978.2 ± 1.8	68325/13304
Protogine Zone dolerites		
Rumperöd	1215 ± 5	62467/13966
Bjärehalla	1221 ± 16	62004/13770
Central Scandinavian Dolerite Group (CSDG)		
Älvdalsåsen	1264.3 ± 1.1	67696/13920
Ämån	1271.1 ± 0.8	68009/14419
Tuna dikes and age equivalents in Dalarna		
Glysjön	1461.2 ± 1.2	68238/13676
Gällsjön	1461.2 ± 1.5	68212/14317
Bunkris	1461.7 ± 1.0	68152/13749
Värmland dolerites		
Ölme	1569 ± 3	65832/14007
Breven–Hällefors dolerites		
Rejmyre	1595 ± 3	65227/15149

matic systems, much greater than the 3–4 epsilon units observed. From this, we conclude that most of the contamination must have occurred in or close to the source region for dolerite magmas. This is also corroborated by the absence of significant correlation between bulk chemical indices of crustal contamination with the isotopic ratios, as observed by Patchett et al. (1994). The fact that Subjotnian dolerite suites (e.g. at c. 1.5 and 1.46 Ga) in Finland and Russia show overall more negative isotope signatures than coeval dolerite suites in Sweden might indicate a stronger influence of contamination by Archaean crust there. The strongly negative ϵ_{Nd}-values of 1.53–1.50 Ga Subjotnian dikes in central Sweden (RDCS in Fig. 4) suggest that these magmas were derived from an Archaean-enriched mantle other than that from which magmas of CSDG were extracted later, or that significant Archaean material contaminated those melts during dolerite intrusion.

A possible scenario proposed by Patchett et al. (1994) for explaining the dominating moderately positive isotope signatures of Proterozoic mafic dikes in Sweden involves enrichment of lithospheric mantle by melts and fluids derived from older sediments that accumulated on the oceanic plate and were brought down to mantle depths in subduction zones. Assuming that subduction was the dominating process affecting local mantle through most of the Palaeoproterozoic, the most likely crustal material available for recycling into subduction zones would be juvenile Svecofennian metaignous crust and metasediments. Repeated subduction-heating-melting-uplift-erosion-sedimentation cycles should leave behind a geochemically and isotopically relatively homogenous mantle that with ϵ_{Hf} and ϵ_{Nd} values lower than normal-depleted mantle (cf. Hergt et al. 1989). Subduction-related recycling of older material mixed with dominantly juvenile material has been proposed as a mechanism for explaining isotopic patterns for other mafic rocks in the areas of the presently investigated dike swarms, and for 1.8 Ga rocks further east (e.g. Eklund et al. 1998; Andersson et al. 2005). Studies elsewhere that reached similar conclusions include those on the Superior Province in North America (e.g. Corfu and Stott 1993), the Proterozoic Ketilidian terrain of Greenland (Patchett and Bridgwater 1984), and Phanerozoic terranes in the Canadian Cordillera (Samson et al. 1989). The $^{147}Sm/^{144}Nd$ ratios in the investigated dolerites range between 0.1182 and 0.1673. This is lower than the chondritic ratio of 0.1967, indicating enrichment of LREE. Similar low ratios for 1.9–1.2 Ga basic rocks in Scandinavia were observed by Claesson (1987). He deduced that if 15–30% partial melting is assumed, recycling of crustal material (with low $^{147}Sm/^{144}Nd$) is a feasible process by which LREE can be enriched in the mantle.

In order to explain the observed positive correlation between initial ϵ_{Nd} and negative Nb anomalies in dolerite samples of CSDG and BDD, Patchett et al. (1994) favoured a mechanism that involved contamination of asthenospheric magma as it ascended through lithospheric mantle. The reasoning was that astenospheric magmas, having no Nb anomaly and initial ϵ_{Nd} typical for a normal-depleted mantle (approximate values are 5 and 6 at 1.26 and 0.96 Ga, respectively), interacted with the lithosphere having negative ϵ_{Nd} composition of about −2 and −2.5. Because much of the Nb in lithospheric mantle is present in refractory spinel, whereas other trace elements reside in less refractory silicates and apatite, less Nb than REE were added to the asthenospheric magmas during chemical interaction with lithospheric mantle. This model is attractive in explaining both the negative Nb anomalies and moderately depleted ϵ_{Nd}signatures.

Regardless of whether magma formation was triggered by partial melting of moderately depleted lithospheric mantle or by contamination of asthenosphere-derived magmas as they migrated through an enriched lithospheric mantle, both models require contamination with crustal material by a process that has caused variably enriched lithospheric mantle domains that could account for the isotopic variations of the dolerite data in Fig. 4. If we assume that major enrichment processes took place in conjunction with early Svecofennian subduction (or during the immediately following TIB episode; cf. Andersson et al. 2005), large volumes of enriched lithospheric mantle sections with $^{147}Sm/^{144}Nd$ values of e.g. 0.16–0.18 may have formed. These would then have had an evolution with time that can account

Table 3 Lu–Hf isotope data

Rock suite	Dolerite	Sample[a]	Lu [ppm]	Hf [ppm]	$^{176}Lu/^{177}Hf$	$^{176}Hf/^{177}Hf$	$\pm 2\sigma$[b]	$\epsilon_{Hf}(0)$	$\epsilon_{Hf}(t)$	Age [Ma]	Age reference
Breven-Hällefors dolerites	Rejmyre	B				0.281850	9	−32.6	+2.9	1596	1
	Breven	WR	0.22	1.54	0.020730	0.282477	9	−10.4	+3.0	1602	2
	Hällefors	WR	0.62	4.01	0.021821	0.282507	4	−9.4	+2.8	1602	12
Värmland dolerites	Ölme	B				0.281855	11	−32.4	+2.4	1565	1
	Bondstorp	WR	0.53	4.86	0.015365	0.282303	10	−16.6	+2.1	1565	12
	Åker A	WR	0.71	6.34	0.015937	0.282292	4	−17.0	+1.1	1565	3,4
	Åker B	WR	0.72	7.35	0.013969	0.282237	14	−19.0	+1.2	1565	3,4
Tuna dolerites and age Equivalents in Dalarna	Gällsjön	B				0.281973	4	−28.2	+4.2	1461	1
	Glysjön A	B				0.281989	8	−27.7	+4.7	1461	1
	Glysjön B	B				0.281983	8	−27.9	+4.5	1461	1
	Bunkris A	B				0.281964	8	−28.6	+3.9	1462	1
	Bunkris B	B				0.281984	7	−27.9	+4.6	1462	1
CSDG	Älvdalsåsen	B				0.282181	3	−20.9	+7.0	1264	1
	Ämån	Z				0.282116	7	−23.2	+5.0	1272	1
	Sorkka	WR	0.34	3.11	0.015739	0.282511	11	−9.2	+5.4	1256	5
	Sorkka A	B				0.282134	4	−22.6	+5.3	1256	5
	Sorkka B	B				0.282146	7	−22.1	+5.7	1256	5
	Sorkka C	B				0.282152	9	−21.9	+5.9	1256	5
	Sorkka, spiked	B	55.80	12000	0.000660	0.282160	9	−21.7	+5.6	1256	5
Protogine Zone dolerites	Steninge-Boda	WR	0.43	5.99	0.010071	0.282430	4	−12.1	+6.8	1220	12
	Röshult	WR	0.70	5.92	0.016837	0.282517	7	−9.0	+4.3	1220	6
	Bjärlöv	WR	0.48	3.92	0.017418	0.282447	39	−11.5	+1.3	1220	12
	Rumperöd	B				0.282122	9	−23.0	+3.9	1215	1,6
	Romeleklint A	WR	0.52	5.81	0.012595	0.282388	6	−13.6	+3.2	1220	12
	Romeleklint B	WR	0.52	5.72	0.012826	0.282377	8	−14.0	+2.6	1220	12
	Bjärehalla	B				0.282127	6	−22.8	+4.2	1221	1
	Taberg A1	WR	0.16	1.45	0.015304	0.282472	12	−10.6	+3.8	1208	7
	Taberg D7	WR	0.04	0.56	0.010915	0.282327	17	−15.7	+2.3	1208	7
BDD	Forserum A	B				0.282209	23	−19.9	+1.5	970	1
	Forserum B	B				0.282206	5	−20.0	+1.4	970	1
	Nilstorp A	B				0.282225	6	−19.4	+2.0	965	1
	Nilstorp B	B				0.282235	21	−19.0	+2.4	965	1
	Särna	B				0.282278	12	−17.5	+4.2	979	1
	Lösen-Fäjö A	B				0.282267	19	−17.8	+2.9	946	1
	Lösen-Fäjö B	B				0.282271	8	−17.7	+3.2	946	1
	Härsjön A	B				0.282264	11	−18.0	+3.1	946	1
	Härsjön B	B				0.282250	8	−18.5	+2.5	946	1
	Nornäs	B				0.282259	4	−18.1	+2.8	947	1
	Falun A	B				0.282324	12	−15.8	+5.0	944	1
	Falun B	B				0.282324	7	−15.8	+5.1	944	1
	Bräkne-Hoby	B				0.282260	6	−18.1	+2.9	948	1
	Karlshamn	WR	0.28	3.56	0.011102	0.282420	7	−12.4	+2.3	954	5
	Karlshamn A	B				0.282222	8	−19.4	+1.7	954	5
	Karlshamn B	B				0.282235	7	−19.0	+2.1	954	5
	Karlshamn C	B				0.282226	8	−19.3	+1.8	954	5
	Karlshamn D	B	9.56	12000	0.000113	0.282230	8	−19.2	+1.8	954	5
	Vångsö	B				0.282294	9	−16.9	+4.1	948	12
Göteborg dolerites	F99022 Tuve-Ö	B				0.282304	11	−16.6	+4.1	935	8
	F99033 Tuve-V	B				0.282247	9	−17.3	+3.4	935	8
	F99003 Heden	WR	0.16	6.21	0.003605	0.282385	7	−13.7	+4.8	935	12
Hunneberg	Coarse, F01-013	WR	0.53	7.23	0.010411	0.282679	5	−3.3	+0.9	282	9
	Coarse, F01-013	B				0.282598	23	−6.1	±0.0	282	9
	Fine	WR	0.39	4.96	0.011166	0.282686	6	−3.1	+1.0	282	9
Skåne dolerites	Arild	WR	0.34	3.85	0.012420	0.282906	6	+4.7	+8.6	284	10
	Mölle	WR	0.35	4.12	0.012063	0.282913	5	+5.0	+9.4	313	10
	Skrylle	WR	0.31	3.54	0.012249	0.282903	5	+4.6	+7.8	284	12
	Horshallen	WR	0.57	4.22	0.019150	0.282459	4	−11.1	−8.4	284	12
Västergötland dolerites	Gaddesanda	WR	0.29	2.36	0.017466	0.282411	8	−12.8	+0.8	1300	12
	Haregården	WR	0.64	7.06	0.012959	0.282310	6	−16.3	+1.1	1300	12
	Lunden	WR	0.61	4.97	0.017465	0.282419	8	−12.5	+1.1	1300	11

$\lambda^{176}Lu$ 1.867 exp^{-11} $year^{-1}$ (Scherer et al. 2001; Söderlund et al. 2004b), CHUR parameters by Blichert-Toft and Albarède (1997). (1) This study, (2) J. Connelly (personal communication), (3) Ask (1996), (4) Söderlund et al. (2004a), (5) Söderlund et al. (2004b), (6) Johansson and Johansson (1990), (7) Larsson and Söderlund (in review), (8) Hellström et al. (2004), (9) Priem et al. (1968), (10) Klingspor (1976), (11) Söderlund et al. (in review) (12) Not dated, age estimated indirectly

[a] *B* Baddeleyite, *WR* whole rock

[b] Errors shown for isotopic composition are two standard errors of the mean for mass spectrometer runs

Table 4 Sm–Nd isotope data

Rock suite	Dolerite	Sm [ppm]	Nd [ppm]	$^{147}Sm/^{144}Nd$	$^{143}Hf/^{144}Nd$	$\pm 2\sigma^b$	$\epsilon_{Nd}(0)$	$\epsilon_{Nd}(t)$	Age [Ma]	Age reference
Breven-Hällefors dolerites	Breven	2.15	8.69	0.1497	0.512177	6	−9.0	+0.7	1602	2
	Hällefors	6.21	23.59	0.1590	0.512274	8	−7.1	+0.6	1602	12
Värmland dolerites and age equivalents	Bondstorp	6.56	30.29	0.1309	0.511975	6	−12.9	+0.3	1565	12
	Romeleklint A	10.21	52.20	0.1182	0.511958	6	−13.3	+2.5	1565	12
	Romeleklint B	10.17	51.84	0.1186	0.511969	7	−13.1	+2.7	1565	12
	Åker A	9.20	42.48	0.1309	0.511961	7	−13.2	±0.0	1565	3,4
	Åker B	9.39	43.38	0.1308	0.511937	7	−13.7	−0.4	1565	3,4
CSDG	Rotsjö	3.78	15.04	0.1519	0.512433	8	−4.0	+3.2	1260	12
	Skutskär	7.52	29.41	0.1547	0.512444	8	−3.8	+3.0	1260	12
	Åstön	4.4	17.49	0.1521	0.512436	13	−3.9	+3.3	1260	12
	Draget	3.88	15.07	0.1557	0.512453	8	−3.6	+3.0	1260	12
Protogine Zone dolerites	Steninge-Boda	8.78	38.93	0.1364	0.512319	7	−6.2	+3.2	1220	12
	Röshult	7.67	30.74	0.1509	0.512367	7	−5.1	+1.5	1220	12
	Bjärlöv	5.54	22.10	0.1515	0.512342	9	−7.9	−1.1	1220	12
	Taberg-gabbro	3.79	18.99	0.1206	0.512001	7	−12.4	−0.6	1208	7
	Taberg Ore	0.63	3.14	0.1207	0.512008	8	−12.3	−0.5	1208	7
Göteborg dolerites	F99003 Heden	7.20	32.08	0.1358	0.512454	6	−3.6	+3.7	935	8
Hunneberg	Grov, F01-013	9.49	43.00	0.1335	0.512406	6	−4.5	−2.2	282	10
	Fin	7.06	31.90	0.1338	0.512417	8	−4.3	−2.0	282	10
Skåne dolerites	Arild	5.59	22.34	0.1513	0.512804	6	+3.2	+4.9	284	10
	Mölle	5.72	22.50	0.1538	0.512798	7	+3.1	+4.8	313	10
	Skrylle	4.72	18.45	0.1546	0.512771	7	+2.6	+4.1	284	10
	Horshallen	5.69	22.25	0.1546	0.512224	14	−8.1	−6.5	284	12
Västergötland dolerites	Gaddesanda	3.35	12.45	0.1626	0.512254	7	−7.5	−1.9	1300	12
	Haregården diabase	10.66	48.22	0.1337	0.512102	7	−10.5	±0.0	1300	12
	Haregården granulite	11.56	52.23	0.1337	0.512079	8	−10.9	−0.4	1300	12
	Lunden	5.65	20.44	0.1673	0.512303	7	−6.5	−1.6	1300	11

Isotope ratios were normalized to $^{146}Nd/^{144}Nd = 0.7219$. CHUR Sm-Nd paramaters are $^{147}Sm/^{144}Nd = 0.1966$ and $^{143}Nd/^{144}Nd = 0.512638$ (Jacobsen and Wasserburg 1980).

[a]Errors shown for isotopic composition are two standard errors of the mean for mass spectrometer runs

[b]Reference numbers refer to those in Table 3

for the Subjotnian dolerites in southeastern Finland, the Jotnian basalts, the Västergötland dolerites, and the Protogine Zone dolerites. Other lithospheric mantle sections may have become less enriched, with $^{147}Sm/^{144}Nd$ values of e.g. 0.18–0.20, and account for most of the variation in the BDD and CSDG.

Tectonic implications

We notice a close temporal relationship between the emplacement of early Mesoproterozoic mafic dolerites and 1.65–1.50 Ga rapakivi magmatism in south Finland, central Sweden and in the Baltic countries. The age of the Breven-Hällefors dike swarm, located east of Lake Vättern (Fig. 1), bridges the gap between the 1.65–1.61 and the 1.59–1.56 Ga rapakivi suites further east. This swarm is ~35 Ma older than the Värmland dolerites in the west and coeval intrusions along the PZ in the south (Fig. 1). The latter intrusions may represent the first episode of extensional movement along the PZ, and fall in the same age range as numerous rapakivi intrusions and dolerites exposed in SW Finland (Suominen 1991). 1.53–1.50 Ga old dike swarms and A-type complexes in central Sweden (Andersson et al. 2002) and Poland (Dörr et al. 2002) may also indicate distant convergent margin processes operating simultaneously from the south, as has been proposed for the earlier (1.85–1.75 Ga) TIB-development in the same area (Högdahl et al. 2004).

The 1,462 Ma Bunkris dike may have critical bearing on the timing of Mesoproterozoic basin formation, sed-

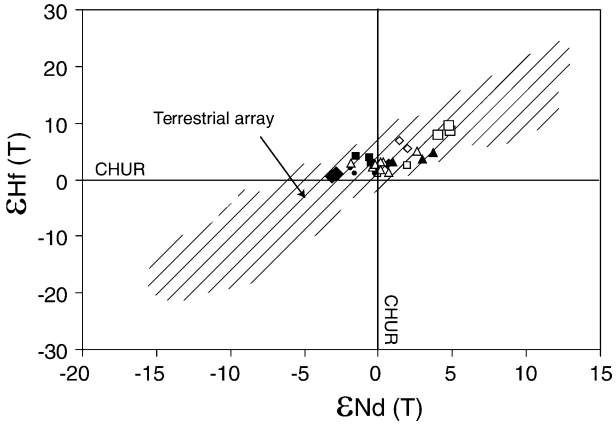


Fig. 5 Initial ϵ_{Hf} versus ϵ_{Nd} for dolerite samples for which both Hf and Nd isotope compositions have been determined. Initial ϵ_{Hf} determined by analysis of baddeleyite (BDD and CSDG) or whole rock samples (Palaeozoic samples) or both baddeleyite and whole rock (all other dolerite generations). Complementary Nd data are from Patchett et al. (1994), Johansson and Johansson (1990), Larsson and Söderlund (in review) and F. Hellström (Göteborg, personal communication). Array for crustal samples defined by striped field from Vervoort et al. (1999). *Symbols* for different generations of dolerites follow Fig. 4

imentation and basaltic volcanism in the central part of the Fennoscandian Shield. The Bunkris dolerite is presumably a feeder dike to the Öje basalt (cf. Nyström 2004), which forms ~100-m thick conformable layer in the 800-m thick Jotnian Dala sandstone sequence. This would imply that at least the lower part of the sedimentary succession was deposited between c. 1.70 Ga (age of basement) and 1.46 Ga. Dolerites and basaltic intercalations that belong to Jotnian formations are scattered over large areas in the Fennoscandian Shield (Nyström 2004; Rämö et al. 2004). In the Nordingrå and Satakunta areas Jotnian sediments rest partly on c. 1.58 Ga rapakivi intrusions, in the Lake Ladoga area on 1.53 Ga rapakivi granite and in the Gävle area on 1.50 Ga rapakivi intrusions (Amantov et al. 1996; Andersson 1997a). In the Dala area of south central Sweden, in addition to the Bunkris dike, the NE-trending swarm of Tuna dolerites are associated and partly mingled with c. 1.47 Ga quartz-feldspar porphyries and the small alkaline Noran intrusion (Claesson and Kresten 1997). Present data suggest that extensive basin formation and associated mafic magmatism was the dominating geology in interior parts of Fennoscandia around 1.45 Ga.

The overall characteristics with respect to timing, regional occurrence and geochemical characteristics of the ~1,595, ~1,568 and the 1,461–1,462 dolerite suites are in favour of dike intrusions in a back-arc setting, and thus in favour of previous proposed tectonic models for Proterozoic Fennoscandia (e.g. Åhäll et al. 2000; Karlstrom et al. 2001; Pesonen et al. 2003; Andersen et al. 2004). The geology of the central part of the Fennoscandian was probably controlled by alternating regimes of compression and extension as response to advancing or retreating subduction zone in the west (see Rivers and Corrigan 2000).

The Central Scandinavian Dolerite Group have less enriched initial isotope signatures (best seen in Hf data Fig. 4, and U. Söderlund, unpublished data) than older mafic rocks in central Fennoscandia. This characteristic might indicate magma production from a mantle source different to that from which the older "subduction-related" mafic magmas were extracted, and that back-arc extension is unrelated to the emplacement of CSDG. This premise corroborates earlier inferences of a linkage to the coeval great Mackenzie dike swarm in North America and the Gardar dikes in Greenland, as well as a contiguous Laurentia and Baltica before the Grenvillian-Sveconorwegian orogeny (e.g. Buchan et al. 2001; Ernst and Buchan 2001; Elming and Mattsson 2001). New, unpublished baddeleyite U–Pb ages (U. Söderlund) reveal E–W chronological zoning of CSDG, i.e. in a direction opposite to that expected if CSDG was directly involved in rifting between Baltica and Greenland. The new data thus support rotation between Baltica and Laurentia after 1.25 Ga (see Romer 1996 and Bingen et al. 1998).

U–Pb data of BDD indicate two pulses of dolerite intrusion at c. 945 and c. 965–970 Ma. The ~30 Ma age difference is almost the same as between Breven-Hällefors and Värmland dolerites, or between CSDG and Protogine Zone dolerites. As shown in Fig. 1, the BDD follows exactly the eastern outline of the Sveconorwegian orogeny in SW Sweden and there is a clear temporal overlap between exhumation (970–930 Ma) of high-grade metamorphic rocks west of the PZ and dolerite intrusion to the east (Söderlund et al. 2004a). Thus, both the older and younger BDD intruded after the main Sveconorwegian compressional stage (see also Patchett and Bylund 1977). The exception is the northernmost BDD, the 978 Ma old Särna dike, which intruded about 10 Ma before Sveconorwegian high-grade metamorphism in SW Sweden.

Conclusions

- Intrusion of dolerite dikes and sills in central Fennoscandia took place repeatedly at ~1,595, ~1,568, 1,461–1,462, 1,256–1,271, 1,215–1,221 and 946–978 Ma. The favoured tectonic model for the tree oldest generations implies dolerite intrusion during discrete events of extension behind a long-lived active continental margin situated along the western (and possibly southern) margin of the Fennoscandian Shield. This evolution started with the Svecofennian orogeny (1.95–1.86 Ga) and continued throughout most of the Mesoproterozoic.
- Comparison of U–Pb baddeleyite ages with previous Rb–Sr and Sm–Nd ages indicates that the latter ages are too young. This probably reflects isotopic disturbance related to alteration in the presence of a fluid at moderate temperatures. Only Rb–Sr data of alteration-free biotite may be useful for dating mafic rocks.
- Initial Hf and Nd isotopic data of Proterozoic dolerites yield isotope signatures corresponding to mildly depleted mantle. Different lithospheric mantle sections with individual degrees of enrichment separate isotopic evolutions through time. Some dolerite generations may represent depleted asthenospheric magmas that partly interacted with such enriched lithosphere.

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